Writing NPU quant kernels: SIMT vs PTO vs ASC

Code size and simulated performance of three tilelang vector frontends implementing one group-wise quantization algorithm.

- 22 kernel variants, each implemented in ASC, PTO and SIMT plus a PyTorch reference: 88 kernels, all passing `python -m harness.check`
- msprof-simulated VF cycles for all 66 NPU kernels
- tilelang 0.1.15, ptoas vmi 0.1.9, CANN 9.2.0-beta.2, SoC `Ascend950PR_9599`
- Single core, `T.Kernel(1)`. Only the `T.SimdVF` / `T.SimtVF` region differs between the three implementations of a variant; tiling, DMA and buffer allocation are identical.

tilelang frontend comparison

Group-wise quantization, in PyTorch:

```
grouped = x.float().view(m, k // group_size, group_size)
amax = grouped.abs().amax(dim=-1).clamp(min=E4M3_CLAMP_MIN)
sf = amax / E4M3_MAX
q = (grouped * (E4M3_MAX / amax).unsqueeze(-1)).view(m, k).to(torch.float8_e4m3fn)
```

frontend	abstraction level	smallest unit of data
ASC (`T.simd` , `S`)	named hardware instruction and distribution mode	one 64-lane vector register
PTO/VMI (`T.vmi` , `V`)	operation with width and segment count as arguments	one logical vector, any width
SIMT (`T.SimtVF` , `T.Parallel`)	per-element expression	one element, held by one thread

`T.SimdVF` / `T.SimtVF` regions range from **6 to 57 lines** across the 22 variants. msprof-simulated VF cycles for the same variant differ by **10-100x between the SIMT and the SIMD implementations**, and by up to 1.6x between the two SIMD frontends.

ASC vs PTO: fewer operations, comparable cycles

PTO expresses the same 22 kernels in **74% of ASC's vector operations** and **89% of its source lines**. Simulated cycles do not follow:

PTO relative to ASC	
vector operations written	74%
`T.SimdVF` source lines	89%
simulated VF cycles	**0.92x mean**, range 0.64x-1.13x

- PTO uses fewer cycles on **9 of 22** variants, more on 6, and is within 2% on 7
- On 4 of those 9 it writes *more* operations than ASC
- Operations written and cycles executed correlate at **Pearson r = -0.09** (n = 22): across this set, source size carries no information about cycle count

Both SIMD frontends are 3-100x faster than SIMT on every variant.

Benchmark setup (and caveats resolved)

Cycle measurement. Sum the `cycles` column of `core0.veccore0_instr_exe.csv` over rows whose `instr` is a VF launch (`VF` for SIMD, `VF_SIMT` for SIMT). The filter keys on `instr`, not `pipe`: every VF row sits on pipe `PUSHQ`, which `MOVEMASK` and `PUSH_PB` also use.

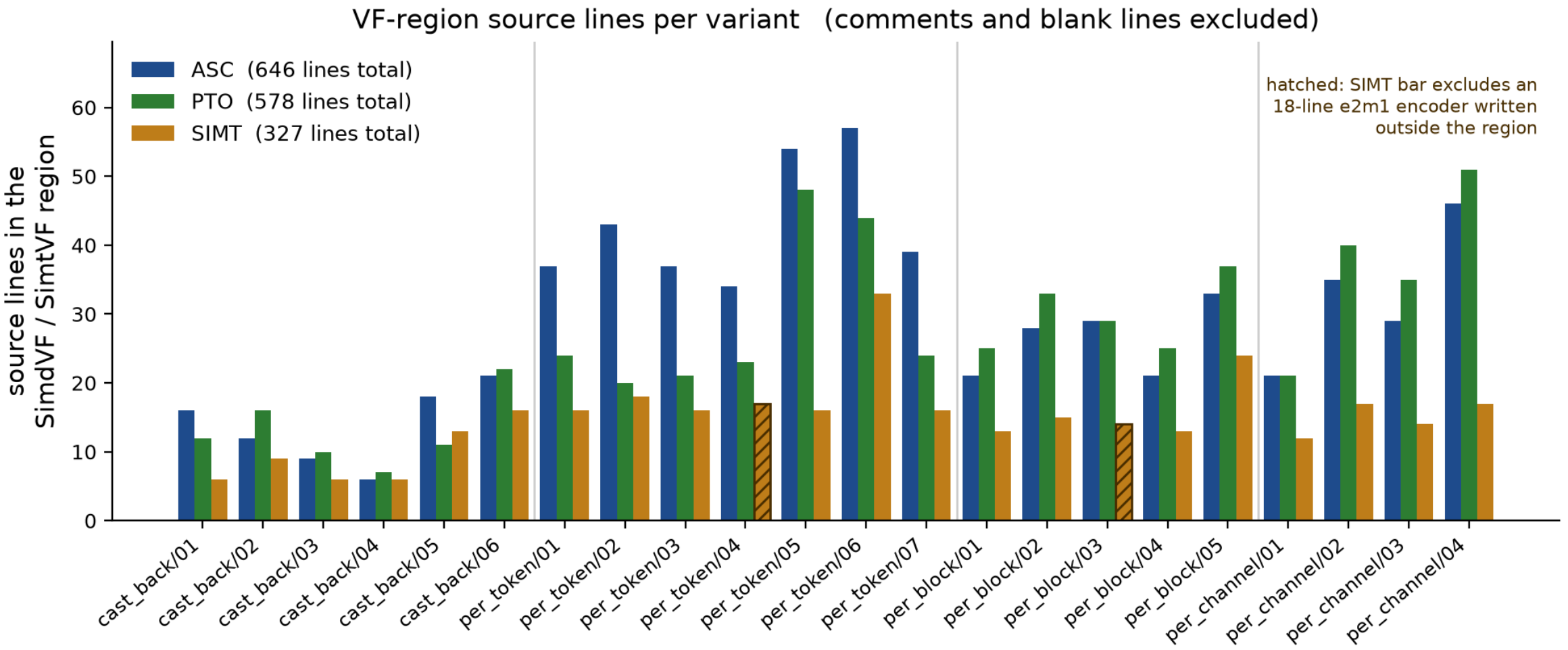
Correctness. Each kernel is compared against `harness/oracle.py`, with an assertion that the result was read back from the device. Cycles are not reported for a kernel that computed a wrong result.

Equal work. VF launch counts agree across the three frontends on every variant except `per_token/05` — ASC and PTO issue 34, SIMT 32, the difference being the gather index SIMT does not build.

Two measurement errors found and corrected. `per_token/06` and `per_block/05` allocated their input with `torch.zeros(device=...)`, an aten fill *kernel*; msprof profiles the first operator, so all three frontends had measured that fill. Both are now driven by `tools/vf_drive.py`, and `tools/vf_trace_audit.py` checks all 66 instruction logs for the arithmetic a quant kernel must contain.

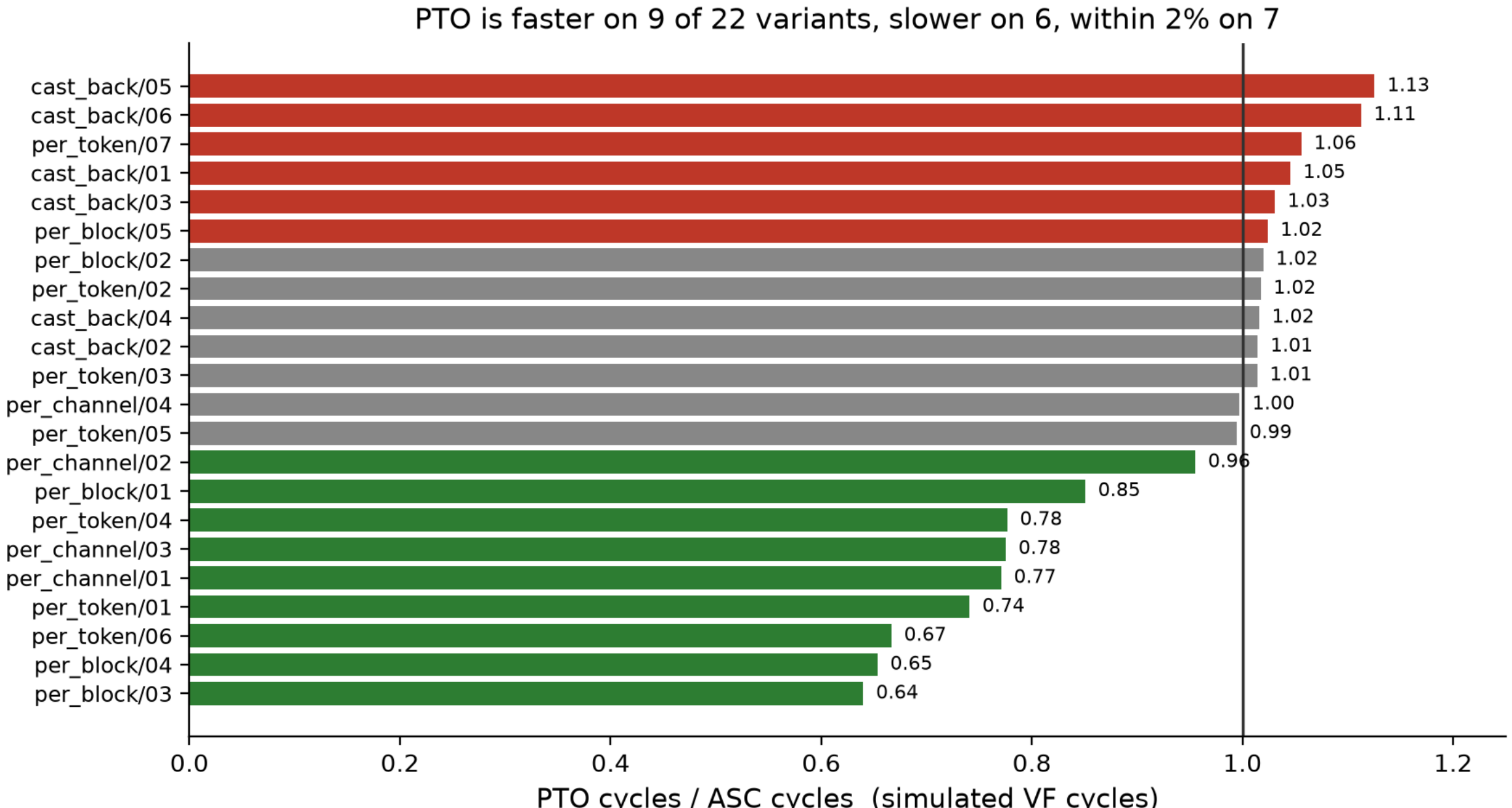
Reading the code fragments. Each is cut verbatim from a kernel the harness passes, and re-checked on every build. Within a variant the three fragments cover the same step; the SIMD fragments' enclosing `for ... in T.serial(...)` tile loop is elided and the SIMT fragment's `T.Parallel` header kept, because that header is what replaces it.

Source size: VF-region lines, all three frontends

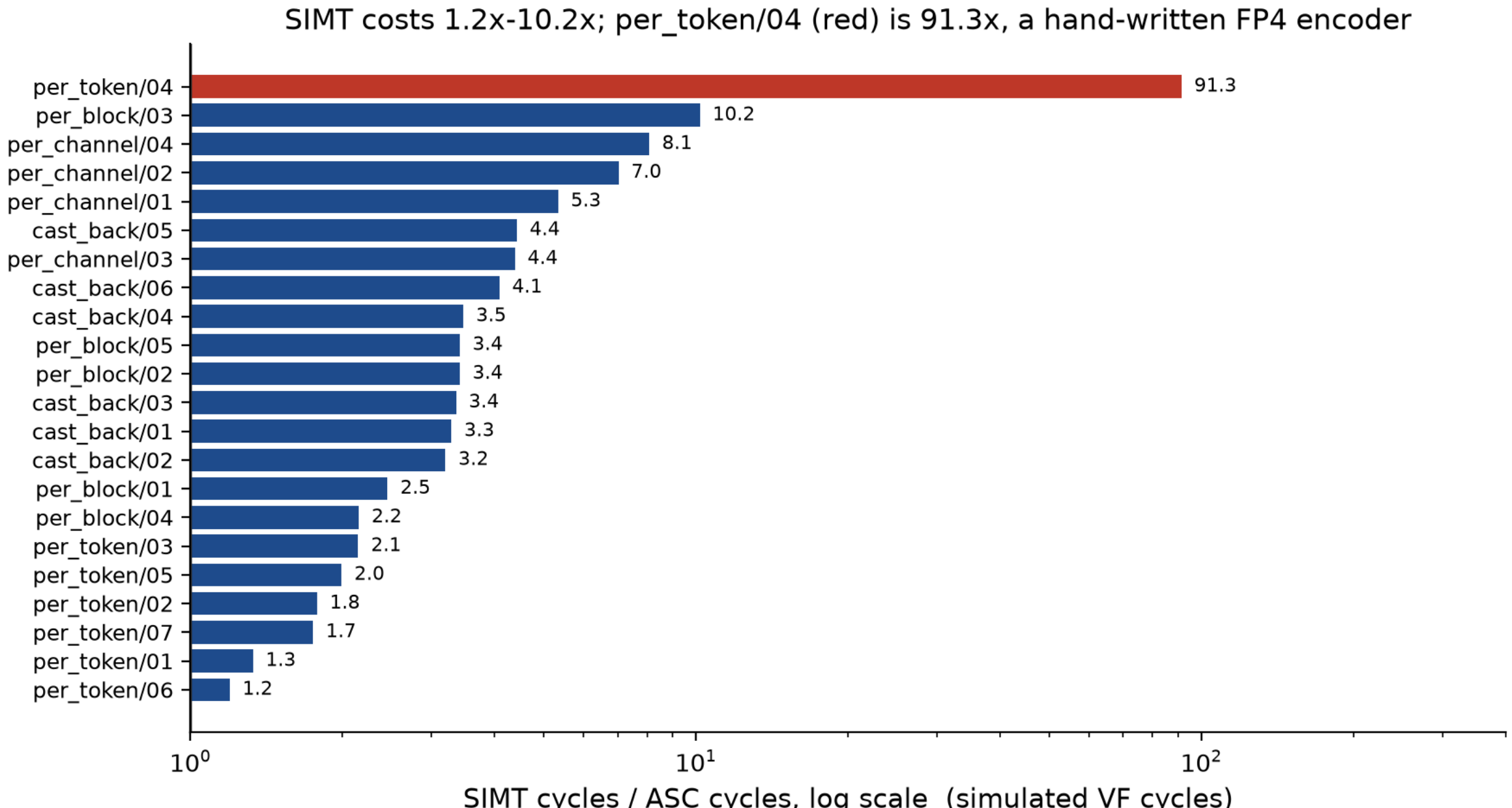


Totals **646 / 578 / 327** lines — PTO 89% of ASC,
SIMT 51%. PTO is *longer* than ASC on 11 of the 22: its
saving is concentrated in the segmented-reduction variants (A2) and spent again
on `size=` and mask operands elsewhere (B3). SIMT is shorter on
21 of 22 and ties on the other.

PTO vs ASC: simulated cycles per variant



SIMT vs ASC: simulated cycles per variant



Walkthrough: nine variants, four kinds of layout mismatch

The algorithm is the same in all 22 variants. What changes is where its quantities sit relative to the 64- or 128-lane vector register.

	variant	code pattern	what it decides
A reduce	`per_channel/01`	one channel per lane	nothing to simplify
	`per_token/01`	4 groups per register	PTO's clearest win
	`per_block/01`	32x32 tile to a scalar	one fold, not two levels
B scale	`cast_back/01`	one scalar across lanes	SIMT's first real cut
	`per_token/02`	exponent, any width	PTO's width argument
	`cast_back/02`	nothing to factor out	PTO's operands cost lines
C output	`per_channel/02`	interleave two registers	SIMT drops an instruction
	`per_token/05`	transposed store = gather	SIMT's biggest cut
D convert	`per_token/04`	float32 -> e2m1	SIMT's 91.3x

The other 13 variants compose these patterns and are measured in the tables at the end. Long form: companion report, sections 2-8.

A1 `per_channel/01`: the reduction axis is already the lane axis

One scale per channel: the reduction runs down the token axis, so a channel stays in one lane and no frontend performs a cross-lane operation.

ASC, accumulating across the 32 rows:

```
acc = S.alloc_local((1,), T.float32)
# seeding with the clamp floor makes the clamp free
acc[0] = S.vdup(E4M3_CLAMP_MIN, T.float32)
for row in T.serial(group_tokens):
    v = S.vabs(S.vcvt(S.vld(x_ub[row, col], dist="UNPK_B16"),
                      T.float32))
    # lane c holds the max over tokens seen so far
    acc[0] = S.vmax(acc[0], v)
S.vsts(sf_ub[col], S.vdiv(acc[0], qmax))
S.vsts(inv_ub[col], S.vdiv(qmax, acc[0]))
```

PTO — the same, with mask and width operands:

```
acc = V.alloc_local((1,), V.vreg(LANES, T.float32))
# seeding with the clamp floor makes the clamp free
acc[0] = V.vbrc(T.float32(E4M3_CLAMP_MIN), size=LANES)
for row in T.serial(group_tokens):
    v = V.vabs(V.vcvt(V.vload(x_ub[row, col], size=LANES),
                      "float32"), mask)
    # lane c holds the max over tokens seen so far
    acc[0] = V.vmax(acc[0], v, mask)
V.vstore(V.vdiv(acc[0], qmax, mask), sf_ub[col])
V.vstore(V.vdiv(qmax, acc[0], mask), inv_ub[col])
```

A1 `per_channel/01`: equal source, unequal cycles

SIMT gives one thread each channel and loops down the 32 rows — the tier's one scanned reduction (A2), because 128 channels already fill the region:

```
for c in T.Parallel(hidden):
    amax = T.alloc_var(T.float32)
    amax = E4M3_CLAMP_MIN
    for row in T.serial(group_tokens):
        amax = T.max(amax,
                     T.abs(T.Cast("float32", x_ub[row, c])))
    sf_ub[c] = amax / E4M3_MAX
    inv_ub[c] = E4M3_MAX / amax
```

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
`per_channel/01`	502	387 (0.77x)	2,677 (5.33x)	21 / 21 / 12

17 and
16 vector operations, 21 lines each:
segment counts and width arguments have nothing to apply to. Cycles differ anyway — 64 lanes against 128.

> **A1** — where the data lines up with the lanes, PTO has nothing to factor out and SIMT nothing to remove. The 0.77x is register width, which no source count can see; it recurs, larger, at A3.

A2 `per_token/01`: a segmented reduction is PTO's clearest win

Each 32-channel group needs its own maximum. An ASC vector reduction produces one value per register, so 4 groups require 4 masked reductions and 4 single-element stores:

```
a0 = S.vabs(S.vcvt(S.vld(x_ub[col], dist="UNPK_B16"), T.float32))
a1 = S.vabs(S.vcvt(S.vld(x_ub[col + LANES], dist="UNPK_B16"),
                   T.float32))
# Each register holds two groups, so reduce half at a time.
S.vsts(amax_ub[group], S.vcmax(a0, mask_low), dist="ONEPT_B32")
S.vsts(amax_ub[group + 1], S.vcmax(a0, mask_high),
       dist="ONEPT_B32")
S.vsts(amax_ub[group + 2], S.vcmax(a1, mask_low),
       dist="ONEPT_B32")
S.vsts(amax_ub[group + 3], S.vcmax(a1, mask_high),
       dist="ONEPT_B32")
```

PTO declares the segment count on one reduction:

```
x = V.vcvt(V.vload(x_ub[col], size=PAIR), "float32")
V.vstore(V.vcmax(V.vabs(x, mask), mask, group=groups_per_pair),
         amax_ub[group])
```

14 vector operations become 5. The matching broadcast on the apply pass collapses from 4 loads plus 2 selects to 1 load.

A2 `per_token/01`: SIMT, and the measured result

SIMT gives one thread each channel and asks the library for the fold:

```
for g, j in T.Parallel(num_groups, group_size):
    x_l[g, j] = T.Cast("float32", x_ub[g * group_size + j])
T.reduce_absmax(x_l, amax_l, dim=1)
```

The group is the fragment's fast axis, so the quant group *is* the reduction axis — the same reshape the CUDA reference makes before its own `T.reduce_absmax(dim=2)`. A clamp pass follows, since a fold cannot be seeded.

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
`per_token/01`	6,176	4,576 (0.74x)	8,204 (1.33x)	37 / 24 / 16

Written instead as one thread per *group* with an inner `T.serial`, the same kernel costs 11.59x. The next slide measures that.

> **A2** — the one pattern where PTO is both shorter and faster:
> 19 operations against ASC's
> 39 at
> 0.74x the
> cycles. Operation count and cycle count favour PTO together on
> 6 of the 22 variants. SIMT is shorter again
> (16 lines) at
> 1.33x.

A2 `per_token/01`: the reduction spelling is worth 8.73x

This is the spelling the tier shipped first — a thread per quant group, the clamp riding along as the accumulator's seed, one line shorter:

```
for g in T.Parallel(num_groups):
    amax_l[g] = E4M3_CLAMP_MIN
    for j in T.serial(group_size):
        v = T.Cast("float32", x_ub[g * group_size + j])
        amax_l[g] = T.max(amax_l[g], T.abs(v))
```

`T.Parallel(num_groups)` has extent 4 at `K=128, group_size=32`, so it keeps 4 of the 128 declared threads busy and issues its inner body 32 times over. `harness/probe/simt_reduce_max.py` builds both spellings, checks both against the oracle, confirms they are byte-identical, and measures each:

reduction spelling	VF_SIMT cycles	vs ASC	instructions
one thread per group, inner `T.serial`	71,606	11.59x	36,676
`T.reduce_absmax` — what this tier ships	**8,204**	**1.33x**	**4,036**

The SIMT numbers in this deck are the library call's, in `per_token` and `per_block` alike; `per_channel` still scans, for the measured reason on the caveats slide. Nothing about the arithmetic changed — only the width of a `T.Parallel`.

A3 `per_block`: the same tile, half as many steps

One scalar per 32x32 tile. **ASC** takes 16 steps of 64 lanes:

```
for chunk in T.serial(num_chunks):
    v = S.vabs(S.vcvt(S.vld(flat_x[chunk * LANES],
                            dist="UNPK_B16"), T.float32))
    S.vsts(run_ub[chunk], S.vcmax(v, mask_all),
           dist="ONEPT_B32")
```

PTO reaches 128 float32 lanes, so the tile takes 8 steps of 128:

```
for chunk in T.serial(num_chunks):
    v = V.vabs(V.vcvt(V.vload(flat_x[chunk * LANES],
                              size=LANES), "float32"), mask)
    V.vstore(V.vcmax(v, mask, group=1), run_ub[chunk])
```

`per_block`	`/01`	`/02`	`/03`	`/04`	`/05`
operations, ASC -> PTO	21->22	27->30	28->26	21->22	29->31
cycles, PTO/ASC	**0.85x**	1.02x	**0.64x**	**0.65x**	1.02x

PTO writes more operations on 4 of these 5 and runs fewer cycles on 2 of those 4.

A3 `per_block/01`: SIMT has no tile loop, and folds in one call

A fragment's shape is an index space, so the 32x32 tile can be declared as a single row and reduced in one call — no steps, no partials:

```
for i, j in T.Parallel(bm, bk):
    flat_l[0, i * bk + j] = T.Cast("float32", x_ub[i, j])
T.reduce_absmax(flat_l, tile_l, dim=1)
```

What remains for one thread is three scalar operations, not a loop:

```
for _z in T.Parallel(1):
    tile = T.max(tile_l[0], E4M3_CLAMP_MIN)
    sf_ub[0] = tile / E4M3_MAX
    inv_ub[0] = E4M3_MAX / tile
```

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
`per_block/01`	1,086	924 (0.85x)	2,664 (2.45x)	21 / 25 / 13

> **A3** — more operations per step and half as many steps executed: a static count cannot observe a trip count. SIMT has neither, and needs neither: the two-level shape a GPU programmer writes first — a partial per row, then one thread folding 32 of them — measured 19,276 cycles against 2,664 here.

B1 `cast_back/01`: broadcasting one scalar across a lane range

64 consecutive channels span two scale groups: each lane needs one of two scalars. **ASC** — two broadcast loads and a lane-mask select:

```
lo = S.vld(sf_ub[group], dist="BRC_B32")
hi = S.vld(sf_ub[group + 1], dist="BRC_B32")
scale = S.vsel(lo, hi, mask_low)

scaled = S.vmul(values, scale)
```

PTO — the segment structure declared on the load:

```
scale = V.vload(sf_ub[group], size=LANES, stride=1,
                dist_mode="brc", group=groups_per_strip)

scaled = V.vmul(values, scale, mask)
```

SIMT — an index expression:

```
for i in T.Parallel(hidden):
    o_l[i] = q_l[i] * sf_ub[i // group_size]
```

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
`cast_back/01`	2,105	2,201 (1.05x)	6,908 (3.28x)	16 / 12 / 6

B1 `cast_back/01`: output dtype selection, and the result

The variant also switches output dtype. **ASC** picks a different store instruction; the wider dtype takes the simpler one:

```
if out_dtype == "bfloat16":
    S.vsts(out_ub[col], S.vcvt(scaled, T.bfloat16),
           dist="PK_B32")      # 64 lanes -> 64 x 2 bytes
else:
    S.vsts(out_ub[col], scaled,
           dist="NORM_B32")    # 64 lanes -> 64 x 4 bytes
```

PTO infers the packing from the destination buffer:

```
if out_dtype == "bfloat16":
    V.vstore(V.vcvt(scaled, "bfloat16"), out_ub[col])
else:
    V.vstore(scaled, out_ub[col])
```

SIMT emits no dtype branch at all: `T.copy` converts between buffers.

> **B1** — SIMT's source is materially shorter for the first time (16 lines to 6): per-element indexing replaces explicit lane-mask construction, at 3.28x, the 10th lowest SIMT ratio of 22.

B2 `per_token/02`: lane count as a parameter

The scale is rounded up to a power of two by rewriting the float32 exponent field. **ASC** fixes the lane count in every type name (`"uint32x64"`):

```
one = S.vdup(1, T.uint32)
bits = T.reinterpret(S.vmuls(clamped, inv_qmax), "uint32x64")
# ((bits - 1) >> 23) + 1  ==  ceil(log2(v)) + 127
biased = S.vadds(S.vshrs(S.vsub(bits, one), 23), 1)
sf = T.reinterpret(S.vshls(biased, 23), "float32x64")
# 254 - biased == 127 - ceil_exp: the negated exponent
inv = T.reinterpret(
    S.vshls(S.vsub(S.vdup(254, T.uint32), biased), 23),
    "float32x64")
```

PTO takes the width as an argument, so one `@T.macro` body — `compute_scale(amax, lanes)` — serves 4, 64, 128 or 256 lanes:

```
one = V.vbrc(T.uint32(1), size=lanes)
shift = V.vbrc(T.uint32(23), size=lanes)
bits = V.vinterpret_cast(
    V.vmul(clamped, V.vbrc(T.float32(inv_qmax), size=lanes), mask), "uint32")
# ((bits - 1) >> 23) + 1  ==  ceil(log2(v)) + 127
biased = V.vadd(V.vshr(V.vsub(bits, one), shift), one)
sf = V.vinterpret_cast(V.vshl(biased, shift), "float32")
# 254 - biased == 127 - ceil_exp: the negated exponent, still biased.
inv = V.vinterpret_cast(
    V.vshl(V.vsub(V.vbrc(T.uint32(254), size=lanes), biased), shift), "float32")
```

B2 `per_token/02`: the same arithmetic on scalars

SIMT has no vector width to carry. Its first six lines are A2's reduction unchanged; the scale arithmetic is the last four:

```
for g in T.Parallel(num_groups):
    amax = T.max(amax_l[g], E4M3_CLAMP_MIN)

    bits = T.reinterpret("int32", amax / E4M3_MAX)
    biased = ((bits - 1) >> 23) + 1
    sf_l[g] = T.reinterpret("float32", biased << 23)
    inv_l[g] = T.reinterpret("float32", (254 - biased) << 23)
```

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
`per_token/02`	4,608	4,688 (1.02x)	8,197 (1.78x)	43 / 20 / 18

> **B2** — 45 operations against
> 14 is the largest operation-count reduction in
> the set, at a
> 1.02x cycle
> ratio. PTO's gain here is a definition that is written once instead of per width; it is not a speed gain, and reporting only the operation count would have presented it as one.

B3 `cast_back/02`: where PTO's explicitness costs source length

Decoding a UE8M0 exponent byte. **ASC** names the distribution mode:

```
packed = S.vld(sf_ub[strip], dist="BRC_B16")
bits = S.vand(S.vshl(T.reinterpret(packed, "uint32x64"),
                     sf_shift), exp_mask)
# An exponent field with a zero mantissa *is* 2^(e-127).
scale = T.reinterpret(bits, "float32x64")
```

PTO states width, stride, group count and mask:

```
packed = V.vload(sf_ub[strip], size=LANES * 2, stride=1,
                 dist_mode="brc", group=1)
bits = V.vand(V.vshl(V.vinterpret_cast(packed, "uint32"),
                     sf_shift), exp_mask)
scale = V.vinterpret_cast(bits, "float32")
```

SIMT shifts by a per-element amount, which VMI cannot express — `V.vshrs` takes a scalar:

```
for i in T.Parallel(hidden):
    g = i // group_size
    word = T.Cast("int32", sf_ub[g // PACK_FACTOR])
    byte = (word >> (8 * (g % PACK_FACTOR))) & 0xFF
    o_l[i] = q_l[i] * T.reinterpret("float32", byte << 23)
```

> **B3** — the counter-example to B2. With nothing to factor out, operations tie at 12 and PTO runs 16 lines against 12; SIMT writes 9 at 3.20x, the tier's lowest.

C1 `per_channel/02`: M-packed scales require a lane interleave

Output byte `2c` holds m-group `2i`'s exponent for channel `c`, byte `2c+1` holds m-group `2i+1`'s. On a vector unit this is an interleave instruction.

ASC:

```
a = S.vld(sf_rows_ub[0, col], dist="NORM_B8")
b = S.vld(sf_rows_ub[1, col], dist="NORM_B8")
# A uint8 register is 256 lanes, so `lo` already holds the
# interleave of the first 128 elements of each input --
# exactly the 256 output bytes for these 128 channels.
# `hi` interleaves the padding and is discarded.
lo, _ = S.vintlv(a, b)
S.vsts(packed_ub[col * PACK_FACTOR], lo, dist="NORM_B8")
```

PTO — same instruction, with the vector width given explicitly:

```
a = V.vload(sf_rows_ub[0, col], size=BYTE_LANES * 2)
b = V.vload(sf_rows_ub[1, col], size=BYTE_LANES * 2)
# A uint8 register is 256 lanes, so `lo` already holds the
# interleave of the first 128 elements of each input --
# exactly the 256 output bytes for these 128 channels.
# `hi` interleaves the padding and is discarded.
lo, _ = V.vintlv(a, b, byte_mask)
V.vstore(lo, packed_ub[col * PACK_FACTOR])
```

C1 `per_channel/02`: SIMT replaces the interleave with two scalar stores

```
for c in T.Parallel(hidden):
    packed_ub[PACK_FACTOR * c] = sf_rows_ub[0, c]
    packed_ub[PACK_FACTOR * c + 1] = sf_rows_ub[1, c]
```

A `uint8` vector register holds 256 lanes, so the loads above read 128 bytes past the 128 meaningful ones. Both SIMD frontends therefore pad the scratch buffer by a whole tile:

```
sf_rows_ub = T.alloc_shared((PACK_FACTOR, hidden + BYTE_LANES), T.uint8)
```


```
sf_rows_ub = T.alloc_shared((PACK_FACTOR, hidden + BYTE_LANES), T.uint8)
```


A scalar store cannot overrun, so SIMT allocates the exact size:

```
sf_rows_ub = T.alloc_shared((PACK_FACTOR, hidden), T.uint8)
```

C1 `per_channel/02`: the measured cost of removing the interleave

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
`per_channel/02`	849	811 (0.96x)	5,964 (7.02x)	35 / 40 / 17

- SIMT removes one instruction and one padding requirement, using 17 lines against ASC's 35, at **7.02x** the cycles
- PTO is **5 lines longer** than ASC (40 vs 35) at 0.96x the cycles: the explicit `size=` and mask operands cost source length, not time

> **C1** — SIMT's largest structural simplification so far: it deletes an instruction *and* a correctness rule about register width. Both deletions are real; 7.02x is what they cost.

C2 `per_token/05`: column-major scales require a register gather

Lane assignment inside a vector register is fixed, so transposing the scale array means computing, per lane, which source element that lane receives.

ASC builds the index vector from the lane id:

```
lane = S.vci(0, T.int32)
token_of_lane = S.vand(T.reinterpret(lane, "uint32x64"),
                       S.vdup(token_block - 1, T.uint32))
group_of_lane = T.reinterpret(S.vshrs(lane, log2_block), "uint32x64")
S.vsts(idx_ub[0],
       S.vadd(S.vmuls(token_of_lane, SF_STRIDE), group_of_lane),
       dist="NORM_B32")
```

...and gathers through it:

```
S.vsts(sf_out_ub[base, 0],
       S.vgather2(sf_dense_ub[0, base], idx), dist="NORM_B32")
```

C2 `per_token/05`: PTO builds the same index; SIMT changes one store

PTO builds the same index vector. VMI has no scalar-operand `vmul`, so the stride is broadcast first — 10 lines against ASC's 7 — and the gather needs its mask:

```
lane = V.vci(T.int32(0), size=LANES)
mask64 = V.create_mask(LANES, size=LANES)
token_of_lane = V.vand(V.vinterpret_cast(lane, "uint32"),
                       V.vbrc(T.uint32(token_block - 1), size=LANES))
group_of_lane = V.vinterpret_cast(
    V.vshrs(lane, log2_block, mask64), "uint32")
# No scalar-operand vmul in VMI: the stride must be a vector.
stride_v = V.vbrc(T.uint32(SF_STRIDE), size=LANES)
V.vstore(V.vadd(V.vmul(token_of_lane, stride_v, mask64),
                group_of_lane), idx_ub[0])
```

```
V.vstore(V.vgather(sf_dense_ub[0, base], idx, gmask),
         sf_out_ub[base, 0])
```

C2 `per_token/05`: in SIMT the transpose is the store's index

The output buffer is declared with the axes already swapped — groups down, tokens across — while the input is read one token at a time:

```
# Already transposed: one column per token of the current block.
sf_out_ub = T.alloc_shared((num_groups, token_block), T.float32)
```

```
for g in T.Parallel(num_groups):
    amax = T.max(amax_l[g], E4M3_CLAMP_MIN)

    # ceil(log2(amax/448)) from the exponent field, then
    # both scales rebuilt by writing an exponent back.
    bits = T.reinterpret("int32", amax / E4M3_MAX)
    biased = ((bits - 1) >> 23) + 1
    inv_l[g] = T.reinterpret("float32", (254 - biased) << 23)
    # The whole column-major config: a different index.
    sf_out_ub[g, sub] = T.reinterpret("float32", biased << 23)
```

`sub` is the token index, from the `T.serial(token_block)` loop outside the region; `g` is the thread's group. The last line is the whole of the column-major configuration: `sf_out_ub[g, sub]` where the row-major variant writes `sf_l[g]`. Everything above it is B2's body, unchanged — there is no index vector, no gather, and no block of scales held back to be permuted.

C2 `per_token/05`: the largest source-size cut in the set

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
`per_token/05`	5,477	5,448 (0.99x)	10,897 (1.99x)	54 / 48 / 16

- 16 SIMT lines against ASC's 54 is the largest source-size reduction among the 22 variants
- it costs 1.99x the cycles, the 5th lowest SIMT ratio of 22 — this row used to be the sharpest source-size / cycle *trade* in the deck, at 13.74x, before its reduction was re-spelled (A2)
- ASC and PTO are within 1% of each other: a gather is a single hardware operation, so VMI has no segment structure or width to collapse here

> **C2** — one line of SIMT replaces nine of index arithmetic, and the SIMD frontends tie because the hardware already provides the operation. Where the work *is* index arithmetic, SIMT is the shortest by a wide margin and no longer pays much for it.

D1 `per_token/04`: no FP4 convert is reachable from a `SimtVF` region

The hardware convert exists and all three frontends run on it. ASC and PTO name it — `S.vcvt(v, T.float4_e2m1fn)`, `V.vcvt(v, "float4_e2m1fn")` — so their work is limited to pre-rounding to bfloat16 with a sticky bit. Nothing in the SIMT surface reaches it: inside a `T.SimtVF` region the arithmetic is plain operators on scalars, and `T.copy` into an FP4 buffer does not compile (`harness/probe/simt_limits.py`).

ASC pairs two registers to deinterleave them, then converts:

```
low, high = S.vdintlv(T.reinterpret(q0, "uint16x128"),
                      T.reinterpret(q1, "uint16x128"))
odd = S.vor(high, S.vmin(low, one_u16))
S.vsts(q_ub[col],
       S.vcvt(T.reinterpret(odd, "bfloat16x128"),
              T.float4_e2m1fn), dist="PK4_B32")
```

PTO uses `V.vunzip`, which splits one value instead of pairing two. It is a `@T.macro`, so the call site is where ASC's three inline statements would be:

```
low, high = V.vunzip(x, "uint16")
one = V.vbrc(T.uint16(1), size=lanes)
return V.vinterpret_cast(V.vor(high, V.vmin(low, one)), "bfloat16")
```

```
odd = to_bf16_round_odd(q, PAIR)
V.vstore(V.vcvt(odd, "float4_e2m1fn", rounding="R"), q_ub[col])
```

D1 `per_token/04`: SIMT encodes e2m1 in software

So the SIMT kernel declares its output `uint8` and extracts the float32 fields by hand — the first 6 lines of an 18-line encoder covering sign, exponent, subnormal renormalization, guard/round/sticky and saturation:

```
bits = T.reinterpret("int32", v)
# bit 3 of (bits >> 28) is bit 31 of bits: the sign nibble, without
# depending on whether >> sign-extends.
sign_nib = (bits >> 28) & 0x8
exps = (bits >> 23) & 0xFF
mant = bits & 0x7FFFFF
```

measured	cycles ASC	cycles PTO	cycles SIMT	VF lines A/P/S
`per_token/04`	6,160	4,784 (0.78x)	562,280 (91.28x)	34 / 23 / 17

The result is bit-exact against the PyTorch reference packer. `per_block/03` carries the same encoder at 16.77x.

D1 `per_token/04`: 91.3x decomposes into instructions and divergence

factor	SIMT `per_token/04`	other SIMT `per_token` variants
instructions issued	243,130	4,036-5,828
of those, branches	44,835	0
of those, predicate sets	22,788	0

The second factor is branch divergence. `T.if_then_else` on a data-dependent per-element condition lowers to a branch rather than a predicated select, and this is the only variant whose trace carries a large number of `SIMT_BRANCH` and `SIMT_START_DVG` markers.

Rewriting the encoder's three conditionals as mask arithmetic (`m = -T.Cast("int32", cond)`, then `(a & m) | (b & ~m)`) issues **37% fewer instructions** and runs at **0.48x the cycles** — 269,370 against 562,280, or **43.7x ASC instead of 91.3x**. Both spellings are in the kernel behind a `branch_free` flag and the harness asserts their outputs are byte-identical.

> **D1** — an instruction the frontend cannot name is not a style difference. Of the 91.3x, a factor of 2.09x is a code-generation gap a GPU compiler would not have, since it predicates such a conditional into a select; the remaining 43.7x is a software encoder standing in for one convert instruction.

Measured data (1/2): `cast_back`, `per_token`

variant	ops A/P	lines A/P/S	cyc A	cyc P	cyc S	P/A	S/A
`cast_back/01`	9/7	16/12/6	2,105	2,201	6,908	1.05	3.28
`cast_back/02`	12/12	12/16/9	2,201	2,233	7,033	1.01	3.20
`cast_back/03`	8/6	9/10/6	2,080	2,144	6,969	1.03	3.35
`cast_back/04`	6/6	6/7/6	2,016	2,048	6,983	1.02	3.46
`cast_back/05`	16/8	18/11/13	2,296	2,584	10,152	1.13	4.42
`cast_back/06`	14/14	21/22/16	2,272	2,528	9,275	1.11	4.08
`per_token/01`	39/19	37/24/16	6,176	4,576	8,204	0.74	1.33
`per_token/02`	45/14	43/20/18	4,608	4,688	8,197	1.02	1.78
`per_token/03`	44/15	37/21/16	4,608	4,672	9,893	1.01	2.15
`per_token/04`	37/17	34/23/17	6,160	4,784	562,280	0.78	91.28
`per_token/05`	56/38	54/48/16	5,477	5,448	10,897	0.99	1.99
`per_token/06`	39/30	57/44/33	6,864	4,576	8,204	0.67	1.20
`per_token/07`	31/17	39/24/16	4,864	5,136	8,502	1.06	1.75

`/06` `full` emits `/01`'s kernel, hence the identical SIMT cycle count.

Measured data (2/2): `per_block`, `per_channel`

variant	ops A/P	lines A/P/S	cyc A	cyc P	cyc S	P/A	S/A
`per_block/01`	21/22	21/25/13	1,086	924	2,664	0.85	2.45
`per_block/02`	27/30	28/33/15	918	936	3,130	1.02	3.41
`per_block/03`	28/26	29/29/14	1,625	1,040	16,536	0.64	10.18
`per_block/04`	21/22	21/25/13	1,414	924	3,044	0.65	2.15
`per_block/05`	29/31	33/37/24	918	940	3,130	1.02	3.41
`per_channel/01`	17/16	21/21/12	502	387	2,677	0.77	5.33
`per_channel/02`	28/29	35/40/17	849	811	5,964	0.96	7.02
`per_channel/03`	23/19	29/35/14	930	721	4,073	0.78	4.38
`per_channel/04`	32/33	46/51/17	738	736	5,964	1.00	8.08

`per_block` reduces a 32x32 tile to one scalar; `per_channel` reduces along the token axis. `per_block/02` and `/05`, and `per_channel/02` and `/04`, share SIMT cycle counts for the same reason: the composed variant emits the earlier variant's `T.SimtVF` region unchanged.

Totals across the 22 variants

	PTO vs ASC	SIMT vs ASC
vector operations	74%	not comparable
source lines	89%	**51%**
simulated cycles	**0.92x** (0.64x-1.13x)	3.73x (1.20x-10.18x)

SIMT has no comparable operation count: its arithmetic is plain operators, so the backend selects the instructions.

- PTO uses fewer cycles on 9 of 22 variants, more on 6, within 2% on 7
- The SIMT range excludes `per_token/04`; including it the maximum is 91.28x
- SIMT costs under 6x on 18 of 22 variants and over 10x on 11

Caveats

1. **Simulated cycles, not hardware.** `msprof op simulator`, SoC `Ascend950PR_9599`. No silicon measurement is claimed.
2. **One shape per variant**, mostly `M=32, K=128`. Ratios compare three implementations of one variant and are valid; absolute cycle counts are not comparable between variants.
3. **Single core.** No multi-core scheduling, double buffering or cache hints.
4. **The reductions are spelled for speed; nothing else is.** Every fold in `per_token` and `per_block` is `T.reduce_absmax` (A2, A3); `per_channel` scans, because its fold runs down the token axis where one thread per channel already uses all 128 threads — 1.06x, measured — and because on its two byte-packing variants the library call does not lower. The format work and the `cast_back` family are untuned. A separate SIMT-vs-SIMD suite sweeping several strategies per kernel measured 0.33x-53x, including SIMT faster on some grouped reductions.
5. **Line counts exclude helpers defined outside the VF region**, in all frontends. This understates SIMT by 18 lines on each of the two FP4 variants.

Caveat 6: the declared SIMT thread count

Every SIMT kernel declares `T.SimtVF(threads=128)`. At `K=128, group_size=32` the two passes inside the region have different extents:

```
for g in T.Parallel(num_groups):     extent 4    (the reduction)
for i in T.Parallel(hidden):         extent 128  (the apply)
```

Measured on `per_token/01`, the same kernel built at three declared widths (`harness/probe/simt_thread_count.py`, all three verified against the oracle):

`threads`	simulated cycles	vs 128	correct
32	8,911	1.09x	yes
64	**6,661**	**0.81x**	yes
128	8,204	1.00x	yes

It is worth 19%, and every kernel here declares 128. At `threads=64` this variant lands 1.08x ASC, 8% off. The same sweep on the serial-scan spelling the tier used to ship found all three widths within 4% of each other — the declared width only started to matter once the fold stopped wasting threads. Nothing in this deck is tuned on it: 128 throughout.

Conclusions

SIMT reduces source size where the SIMD frontends move data between lanes. A group broadcast (B1, 16 lines to 6), a lane interleave (C1, 35 to 17) and a register gather (C2, 54 to 16), costing 3.28x, 7.02x and 1.99x cycles — index arithmetic the thread model does not require, plus the 256-lane padding rule.

A `T.Parallel` must be as wide as its data, and that was most of the SIMT column. Handing each fold to `T.reduce_absmax` instead of one thread per quant group took `per_token/01` from 11.59x to 1.33x and `per_block/01` from 17.75x to 2.45x, output unchanged (A2, A3).

PTO simplifies where a segment count or a width replaces repetition (A2, B2: 39 operations to 19 and 45 to 14) **and costs source length where there is nothing to factor out** (B3: 4 lines longer at an identical operation count).

Neither SIMD frontend has anything to simplify where the operation already maps onto the hardware — a channel in one lane (A1), or a scalar scale (`per_block/01`): 17 operations against 16 and 21 lines each.

Where the hardware has an instruction, the frontend that can name it wins by a wide margin. All three run on the same unit; only ASC and PTO have a call reaching the e2m1 convert, and FP4 output (D1) costs SIMT 91.3x — 2.09x of it a code-generation gap (branch divergence), the remaining 43.7x a software encoder standing in for one instruction.

Frontend selection

SIMT — when the kernel is dominated by data placement (packing, transposes, gathers) and development time is worth more than cycles. It is never faster here: the shorter source is bought with cycles.
`per_token/05` *54 lines to 16 at 1.99x* · `per_channel/02` *35 to 17 at 7.02x* · `cast_back/01` *16 to 6 at 3.28x*

PTO/VMI — general production vector code, and anywhere a segment count or a width replaces repetition: 74% of ASC's operations at 0.92x of its cycles.
`per_token/01` *39 operations to 19 at 0.74x* · `per_token/02` *45 to 14 at 1.02x* · `per_block/03` *at 0.64x*

ASC — when an instruction VMI does not expose is required (per-lane variable right shift, which SIMT can also express — B3), when a distribution mode is the objective, or when there is nothing for VMI to factor out.
`cast_back/02` *12 lines against PTO's 16* · `per_channel/04` *46 against 51* · `cast_back/05` *, PTO's worst cycle regression, 1.13x*

On methodology. Source size correlates with simulated cycles at r = -0.09 on this set: a frontend comparison reporting only source size reports one of the two quantities that matter.